

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

**A Comparative Study of Swin Transformer and ResNet50 for
Automated Detection of Age-Related Macular Degeneration Using
Fundus Images**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**A Comparative Study of Swin Transformer and ResNet50 for Automated Detection of Age-Related Macular Degeneration Using Fundus Images**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

It is our pleasure to be indebted to various people, who directly or indirectly contributed in the development of this work and who influenced our thinking, behavior and acts during the course of study.

We express our sincere gratitude to all for proving us an opportunity to undergo IOHE Project as the part of the curriculum.

We are thankful to Dr. Shikha Tuteja for her support, cooperation and motivation provided to us during the training for constant inspiration, presence and blessings.

Lastly, we would like to thank the almighty and our parents for their moral support and friends with whom we shared our day-to-day experience and received lots of suggestions that improve the quality of work.

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ABSTRACT

Age-related macular degeneration (AMD) is a major cause of irreversible loss of central vision and is considered one of the leading retinal diseases affecting elderly individuals worldwide, particularly people above the age of 75 years. The disease gradually damages the macula, the central part of the retina responsible for sharp and detailed vision, thereby affecting daily activities such as reading, driving, and facial recognition. Early diagnosis of AMD is extremely important for preventing severe vision impairment; however, accurate detection remains challenging because retinal image interpretation requires experienced ophthalmologists, and expert medical resources are often limited in many regions.

To address these challenges, this study explores the use of advanced deep learning techniques for automated AMD detection from retinal images. One of the major issues in medical imaging datasets is data imbalance, where healthy retinal images significantly outnumber AMD-affected images. To overcome this limitation and improve model generalization, the Synthetic Minority Oversampling Technique (SMOTE) is employed to generate synthetic samples of the minority AMD class. This helps in reducing bias during training and improves the overall robustness of the classification model.

The study specifically focuses on two powerful deep learning architectures: the Swin Transformer and ResNet50. The Swin Transformer utilizes a hierarchical window-based self-attention mechanism that effectively captures global contextual features from retinal images, making it suitable for identifying subtle disease patterns. On the other hand, ResNet50, a widely used deep convolutional neural network, is capable of extracting detailed local spatial features while maintaining computational efficiency, even when trained on relatively smaller datasets. By combining these approaches, the study aims to enhance the accuracy, reliability, and efficiency of automated AMD detection systems in real-world clinical applications.

Keyword: Age-related macular degeneration, retinal fundus imaging, deep learning, convolutional neural network, Swin Transformer, ResNet50, SMOTE

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1. INTRODUCTION

1.1. Background

Age-related Macular Degeneration (AMD) is a progressive retinal disorder and one of the primary causes of irreversible central vision loss among older adults worldwide. The disease affects the macula, the central part of the retina responsible for sharp and detailed vision. As AMD progresses, patients experience difficulties in daily activities such as reading, driving, and recognizing faces. According to recent global estimates, nearly 196 million people were affected by AMD in 2020, and the number is expected to increase significantly with the growing aging population. The increasing prevalence of AMD has created major healthcare and socioeconomic challenges, highlighting the importance of early diagnosis and timely treatment [1].

Advancements in Artificial Intelligence (AI) and Deep Learning have introduced new possibilities in the field of medical image analysis. Deep learning models, especially Convolutional Neural Networks (CNNs), have demonstrated strong performance in retinal disease detection by automatically learning important features from retinal fundus images. ResNet50 is one such deep CNN architecture that efficiently extracts local spatial features while maintaining high classification accuracy [2]. In addition, Vision Transformer (ViT)-based architectures such as the Swin Transformer have recently gained attention because of their ability to capture global contextual information using self-attention mechanisms. The Swin Transformer further improves efficiency through hierarchical feature extraction and shifted window attention, making it suitable for medical imaging applications [3].

1.2. Problem Statement

Despite the availability of advanced imaging techniques, early diagnosis of AMD remains a significant challenge. Traditional diagnosis relies heavily on manual examination of retinal fundus images by expert ophthalmologists, which is time-consuming, subjective, and dependent on clinical expertise. In many healthcare settings, the shortage of trained specialists further limits the availability of timely screening and accurate diagnosis.

Another major challenge in developing automated AMD detection systems is the issue of class imbalance in medical datasets, where healthy retinal images are more common than AMD-affected images. This imbalance can negatively impact the learning ability and generalization performance of deep learning models. Furthermore, conventional CNN-based models may struggle to capture long-range dependencies and broader contextual relationships present in retinal images.

To address these challenges, this study proposes the use of deep learning models, specifically ResNet50 and Swin Transformer, for automated AMD classification from retinal fundus images. The Synthetic Minority Oversampling Technique (SMOTE) is applied to balance the dataset by generating synthetic AMD samples. The study aims to compare the performance of both architectures using evaluation metrics such as accuracy, precision, recall, F1-score, and AUC to determine a reliable and efficient approach for AMD detection.

2. METHODOLOGY

2.1. Dataset

The ADAM dataset was used in this study. It comprises 400 color fundus images distributed across two classes: AMD (89 images) and non-AMD (311 images). This represents a significant class imbalance, with the Non-AMD class being approximately 3.5 times larger than the AMD class. All images were sourced from clinical fundus photography and represent real-world diagnostic scenarios, including cases of early, intermediate, and advanced AMD.

2.2. Preprocessing

Prior to model training, each image was subjected to the following preprocessing steps:

- Image Resizing: All fundus images were resized to 224×224 pixels to meet the input requirements of both ResNet50 and the Swin Transformer.
- Normalization: Pixel intensities were standardized using ImageNet mean and standard deviation values, ensuring compatibility with both pre-trained models.
- Center Cropping: Circular artifacts and black borders commonly introduced by fundus camera optics were removed using center cropping.
- Quality Filtering: Images with significant blur, glare, or illumination artifacts were removed from the dataset prior to training.

2.3. Data Augmentation

To increase dataset diversity and improve the models' robustness to acquisition variability, offline data augmentation was applied to all images before training. Augmented images were stored physically in separate training, validation, and testing directories. After augmentation, each split contained 1,335 AMD images and 3,110 non-AMD images. The following augmentation techniques were applied:

- Rotation (random angles)
- Horizontal and vertical flips
- Color jitter (hue, saturation, brightness adjustments)
- Gaussian noise and Gaussian blur
- Random cropping

Particularly aggressive augmentation was applied to AMD images in order to increase the diversity of the minority class and reduce the risk of overfitting to a small number of original samples.

2.4. Class Imbalance Handling

Given the significant imbalance between the AMD and Non-AMD classes, the Synthetic Minority Oversampling Technique (SMOTE) was employed as an additional class-balancing strategy [4]. SMOTE operates in feature space rather than image space: high-level feature representations were first extracted from the training images using a lightweight ResNet-based architecture (ResNet20). SMOTE was then applied exclusively to the AMD minority class to generate synthetic feature vectors by interpolating between existing AMD feature representations.

These synthetic feature vectors were integrated back into the training pipeline alongside the real extracted features, providing the models with a more balanced class distribution. This approach was intended to supplement data augmentation and prevent the models from developing a systematic bias toward predicting the non-AMD majority class.

2.5. Model Architectures

Two deep learning architectures were selected for comparison:

- **ResNet50:** ResNet50 is a 50-layer deep convolutional neural network using residual connections for training deep networks. Its skip connections allow gradients to flow cleanly through deep layers, enabling effective training without degradation. ResNet50 is well-suited for learning local spatial features and has established strong benchmarks across medical imaging tasks.
- **Swin Transformer:** The Swin Transformer is a hierarchical vision transformer that uses a shifted-window self-attention mechanism to capture both fine-grained local features and high-level global context. Its hierarchical design allows it to process images at multiple scales efficiently, making it suitable for retinal images where pathological features such as drusen deposits, RPE changes, and subretinal fluid can appear at varying spatial scales.

Both models were initialized with ImageNet-1K pre-trained weights and fine-tuned for binary classification by replacing the final classification layer with a single binary output linear layer.

2.6. Training

The training process for both deep learning models was carried out using the PyTorch framework [5]. The input to the models consisted of 224×224 RGB fundus images that were normalized using ImageNet statistics to ensure consistency during training. A batch size of 32 was used, and the models were trained for a maximum of 60 epochs with an early stopping mechanism based on validation accuracy to prevent overfitting. The Adam optimizer was applied for the ResNet50 model, while the Swin Transformer utilized the AdamW optimizer for improved weight regularization. To enhance convergence, the ReduceLROnPlateau learning rate scheduler was employed, which dynamically adjusted the learning rate according to validation performance. Cross-entropy loss with class weights was used as the loss function to address class imbalance in the dataset [6]. Additionally, a WeightedRandomSampler was implemented to maintain balanced mini-batch composition during training. To further improve generalization and reduce overfitting, regularization techniques such as label smoothing and dropout in the classification head were incorporated.

2.7. Evaluation metrics

The model's performance was assessed through the use of various evaluation metrics such as accuracy, precision, recall, F1 score, ROC-AUC, confusion matrix, and training curves. These criteria for evaluating the model's performance were selected based upon common practice followed in the classification of medical images literature

3. TOOLS AND TECHNOLOGIES

3.1. Programming Language and Framework

Python was used as the primary programming language because of its extensive support for deep learning, medical imaging, and scientific computing libraries.

PyTorch was used for implementing, training, fine-tuning, and evaluating the deep learning models.

Both ResNet50 and Swin Transformer were developed using the PyTorch framework.

3.2. Deep Learning Libraries and Models

- ResNet50: A deep Convolutional Neural Network (CNN) based on residual learning, used for extracting hierarchical local spatial features from retinal fundus images.
- Swin Transformer: A transformer-based architecture that uses shifted-window self-attention mechanisms to efficiently capture both local and global image features.
- ImageNet Pretrained Weights: Both models were initialized with pretrained ImageNet-1K weights and fine-tuned for binary AMD classification.

3.3. Data Processing and Augmentation

- torchvision.transforms: Used for image preprocessing and augmentation operations such as resizing, normalization, flipping, rotation, and color jitter.
- Albumentations: Used for advanced augmentation techniques including Gaussian noise, brightness and contrast adjustment, random cropping, and Gaussian blur [7].
- SMOTE (Synthetic Minority Oversampling Technique): Used for handling class imbalance by generating synthetic minority-class feature samples.
- scikit-learn: Used for preprocessing utilities, model evaluation, and integration of performance metrics [6].

3.4. Training and Optimization

- Adam Optimizer: Used for optimizing the ResNet50 model during training.
- AdamW Optimizer: Used for optimizing the Swin Transformer model with improved weight regularization.
- ReduceLROnPlateau Scheduler: Applied for adaptive learning rate adjustment based on

validation performance.

- WeightedRandomSampler: Used to create balanced mini-batches during training.
- Early Stopping and Label Smoothing: Implemented to reduce overfitting and improve model generalization.

3.5. Evaluation and Visualization

- scikit-learn Metrics: Used for calculating accuracy, precision, recall, F1-score, ROC-AUC score, and confusion matrix analysis.
- Matplotlib and Seaborn: Used for visualizing training curves, confusion matrices, ROC curves, and precision-recall curves.

3.6. Hardware and Dataset

- CUDA-enabled GPU Environment: Used for accelerating deep learning model training and inference.
- ADAM Dataset: Publicly available retinal fundus image dataset consisting of AMD and Non-AMD images used for binary classification.

4. IMPLEMENTATION

4.1. Dataset Preparation

The ADAM retinal fundus image dataset was used for binary classification of AMD and Non-AMD cases.

The dataset initially contained 400 color fundus images with class imbalance:

- 89 AMD images
- 311 non-AMD images

Images were organized into separate training, validation, and testing directories.

4.2. Image Preprocessing

All retinal images were resized to 224×224 pixels to match the input requirements of both ResNet50 and Swin Transformer. Image normalization was performed using ImageNet mean and standard deviation values. Center cropping was applied to remove black borders and unwanted background regions from fundus images. Low-quality images affected by blur, glare, or poor illumination were filtered out to improve dataset quality.

4.3. Data Augmentation

Offline augmentation techniques were applied to increase dataset diversity and improve model generalization.

The augmentation operations included:

- Random rotation
- Horizontal and vertical flipping
- Color jitter
- Brightness and contrast adjustment
- Gaussian noise
- Random cropping

More aggressive augmentation was applied to AMD images to strengthen minority-class representation.

4.4. Class Imbalance Handling

Since the dataset was highly imbalanced, SMOTE-based oversampling was incorporated to improve AMD class representation [4]. Synthetic minority-class feature samples were generated in feature space to reduce prediction bias toward the Non-AMD class.

WeightedRandomSampler and class-weighted cross-entropy loss were also used to create balanced mini-batches during training.

4.5. Model Training

Two pretrained deep learning architectures were fine-tuned for AMD classification:

- ResNet50
- Swin Transformer

The final classification layers of both models were replaced with a binary output layer.

Training configuration included:

- Batch size: 32
- Maximum epochs: 60
- Optimizers: Adam and AdamW
- Learning rate scheduler: ReduceLROnPlateau
- Early stopping based on validation accuracy

Label smoothing and dropout techniques were also applied to reduce overfitting.

4.6. Model Evaluation

Both models were evaluated using multiple performance metrics including:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC score
- Confusion matrix analysis

Training and validation loss curves were analyzed to study convergence behavior and generalization performance.

4.7. Comparative Analysis

The final implementation stage involved comparing the performance of both models on the ADAM test dataset.

Experimental results showed that the Swin Transformer achieved:

- Higher accuracy
- Better AMD recall
- Lower false-negative rates
- Improved ROC-AUC score
- Better generalization capability

ResNet50 achieved reasonable performance but showed comparatively higher overfitting and lower sensitivity toward AMD-positive cases.

5. RESULT

5.1 Classification Performance

Classification performance evaluation was performed for both the Swin Transformer and ResNet50 models using the standard measures. Performance results following the finetuning process are presented in table 1 below.

Model	Accuracy	Precision (AMD)	Recall (AMD)	Macro F1	AUC
Swin Transformer	88.68%	0.81	0.82	0.87	0.919
ResNet50	84.00%	0.74	0.71	0.80	0.855

Table 1: Classification Performance After Fine-Tuning

The Swin Transformer achieved a notably higher overall accuracy of 88.68% compared to ResNet50's 84.00%. More critically for clinical applications, the Swin Transformer demonstrated substantially better AMD recall (0.82 vs. 0.71), meaning it correctly identified a higher proportion of true AMD cases. This is supported by the higher AUC of 0.919 vs. 0.855, indicating superior discriminative power across all decision thresholds.

5.2 Confusion Matrix Analysis

Table 2 presents the confusion matrix comparison between the two models.

Model	True AMD	AMD → Non-AMD (FN)	Non-AMD → AMD (FP)
Swin Transformer	1097	238	265
ResNet50	950	385	339

Table 2: Confusion Matrix Comparison

ResNet50 produced 385 false negatives (AMD cases incorrectly classified as Non-AMD), compared to only 238 for the Swin Transformer. In a clinical screening context, false negatives are particularly costly as they represent AMD patients who would be missed and left untreated. The Swin Transformer's lower false-negative rate therefore represents a meaningful clinical advantage.

5.3 ROC-AUC and Precision-Recall

ROC curve analysis confirms the Swin Transformer's superior class separability, with an AUC of 0.919. ResNet50's AUC of 0.855 is respectable but notably lower, indicating that the Swin Transformer maintains better sensitivity-specificity trade-offs across varying thresholds.

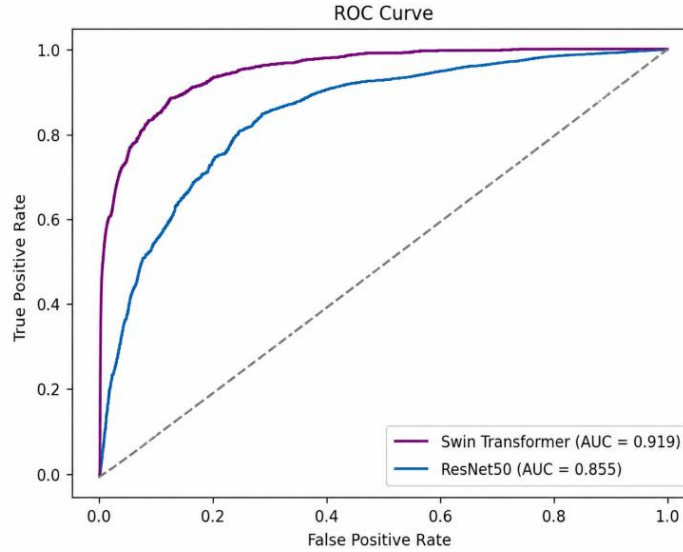


Figure 1. ROC Curve

Precision-recall analysis further supports this finding. The Swin Transformer achieves high precision across nearly all recall values, with an average precision (AP) of approximately 0.90. ResNet50 achieves an AP of 0.896 but shows a steeper precision decline at high recall values, meaning that when the model is pushed to detect more AMD cases, it produces more false positives at a faster rate than the Swin Transformer.

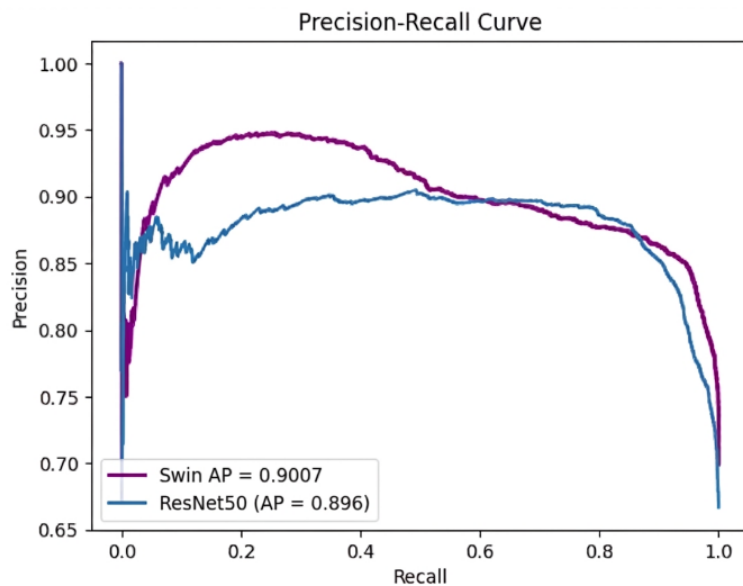


Figure 2. Precision Recall Curve

5.4 Training Dynamics

The Swin Transformer demonstrated stable and consistent convergence. Training and validation loss decreased together, and validation accuracy tracked closely with training accuracy throughout, showing no signs of overfitting. This reflects the model's ability to generalize well from the training data.

ResNet50, by contrast, showed signs of overfitting beginning around epoch 40-50. Training accuracy continued to climb toward 100%, while validation accuracy plateaued at approximately 83-85%. This divergence, accompanied by increasing fluctuations in validation loss, indicates that ResNet50 began memorizing training-specific patterns rather than learning generalized retinal features.

5.5 Key Findings Summary

- The Swin Transformer outperformed ResNet50 on all evaluation metrics, including accuracy, recall, F1-score, and AUC.
- The Swin Transformer's hierarchical shifted-window attention enabled superior detection of spatially distributed retinal biomarkers associated with AMD.
- ResNet50 exhibited overfitting in later training epochs, suggesting limitations in generalizing from augmented retinal image data.
- Both models performed more strongly on the non-AMD class, reflecting the residual influence of class imbalance even after augmentation and SMOTE application.
- The incomplete implementation of SMOTE in ResNet50's training pipeline likely contributed to its weaker AMD recall performance.

6. CONCLUSION AND FUTURE SCOPE

6.1. Conclusion

This study presented a comparative analysis of two deep learning architectures, ResNet50 and Swin Transformer, for automated detection of Age-related Macular Degeneration (AMD) using retinal fundus images. The experimental results demonstrated that the Swin Transformer achieved superior performance compared to ResNet50 across multiple evaluation metrics, including accuracy, recall, F1-score, and ROC-AUC score.

The hierarchical self-attention mechanism of the Swin Transformer enabled effective extraction of both local and global retinal features, leading to improved classification performance and better generalization capability. In contrast, ResNet50, although computationally efficient and effective for local feature extraction, showed limitations in capturing long-range dependencies and exhibited signs of overfitting during training.

The study also highlighted the importance of handling class imbalance in medical imaging datasets. The use of SMOTE-based oversampling and data augmentation techniques contributed to improved minority-class representation and reduced prediction bias toward the majority class.

Overall, the research demonstrates that transformer-based deep learning models have strong potential for automated AMD diagnosis and can support ophthalmologists in early disease detection and clinical decision-making.

6.2. Future Scope

Several directions can be explored to extend and improve upon the findings of this study:

- **Full SMOTE Integration:** Properly implementing SMOTE in ResNet50's training pipeline and re-evaluating performance to measure its isolated impact on minority-class recall and overall generalization.
- **Hybrid CNN-Transformer Architectures:** Developing models that combine ResNet50's efficient local feature extraction with the global attention capabilities of the Swin Transformer. Such hybrid models may leverage the complementary strengths of both paradigms for improved diagnostic accuracy.

- **Expanded and Diverse Datasets:** Training on larger, more diverse datasets that include images from different clinical centers, imaging devices, and patient populations would improve model robustness and generalizability to real-world screening conditions.
- **Multi-Class AMD Staging:** Transitioning from binary classification to multi-class systems capable of distinguishing early, intermediate, and advanced AMD would significantly increase the clinical utility of automated detection systems.
- **Explainability with Grad-CAM:** Integrating gradient-weighted class activation mapping (Grad-CAM) or similar explainability tools would generate visual heatmaps highlighting the retinal regions most influential in the model's decisions. This is essential for establishing clinical trust and supporting ophthalmologist adoption [8].
- **Advanced Regularization Strategies:** Applying stronger regularization techniques such as increased dropout, weight decay, and stochastic depth to ResNet50 to mitigate overfitting and improve its performance at higher training epochs.
- **Clinical Validation:** Deploying the best-performing model in a prospective clinical study to evaluate its performance against specialist ophthalmologist grading under real-world conditions.

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